Tide retrospective: the graded exponential expansion

a Taylor-truncation collection mechanism

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Abstract

The numerator stage of the forward-expansion programme must expand $\exp(-(\sum_{s=1}^{N} q^s a_s + q^N \rho_q))$ to order N in q. The design consult prescribed the log-derivative recursion $P_j = -\frac{1}{j}\sum_{s=1}^{j} s\, a_s P_{j-s}$ as the internal collection mechanism, while noting that the proof itself may use exp's ordinary Taylor remainder. This tide substitutes a cheaper mechanism with the same interface: the *graded exp polynomial* $Q(X) = \sum_{i=0}^{N} \frac{(-1)^i}{i!} A(X)^i$ with $A(X) = \sum_{s=1}^{N} a_s X^s$ — exp's Taylor truncation composed with the exponent polynomial — whose coefficients are the expansion coefficients. With that choice the expansion theorem needs no coefficient identities at all. Three elementary steps close it: strip ρ by the first-order exponential bound $|e^x - 1| \le 2|x|$; apply exp's Taylor remainder at $x = -A(q)$, where $|A(q)| \le q \sum_s |a_s|$ on $(0, 1]$; and observe that the polynomial tail of Q above degree N carries q^{N+1}. Each step is $O(q^{N+1})$ against the target $o(q^N)$.

1 Setting

Stage 4 of the archived implementation order, after the exponent split (stage 3) produced exactly the input shape this lemma consumes: coefficients $a_s = V_s(z)$ and a perturbation ρ_q vanishing at 0^+ pointwise in z. The statement here is scalar — the z-dependence is a family of instantiations, and the function-valued mesoscopic form belongs to the numerator stage, per the consult's own remark. A numerical check preceded the Lean: the polynomial's coefficients at $N = 3$ match the recursion's values ($c_1 = -a_1$, $c_2 = -a_2 + a_1^2/2$, $c_3 = -a_3 + a_1 a_2 - a_1^3/6$), and the remainder over q^N scales as the injected $\rho = \sqrt{q}$.

2 The thing formalised

Definitions: `exponentPoly` (A), `gradedExpPoly` (Q), and `expCorrectionCoeff` ($c_j = Q_{[j]}$), with evaluation lemmas and $c_0 = 1$ (the correction factor tends to one). Helpers: the linear bound $|A(q)| \le (\sum_s |a_s|)\, q$ on $[0, 1]$ (each $q^s \le q$), the vanishing of A at 0^+, and $q^{N+1} = o(q^N)$ transported from Mathlib's two-power comparison at $\mathcal{N}(0)$. The main theorem `exp_graded_expansion`: for any $\rho \to 0$ at 0^+,

$$\exp\Big(-\Big(\sum_{s=1}^{N} q^s a_s + q^N \rho_q\Big)\Big) - \sum_{j=0}^{N} c_j\, q^j = o(q^N) \quad (q \to 0^+),$$

assembled as a three-term telescope, each term big-O of q^{N+1} or of $q^N \rho_q$.

3 Where progress stalled

Three small rounds, all catalogued classes. The norms-both-sides trap in `isLittleO_iff` fired in a new variant: an earlier `abs_of_pos` rewrite had already stripped the norm from the *target's* right side, so the calc had to end at $\varepsilon\, q^N$ rather than $\varepsilon\, |q^N|$ — the rule is to look at what previous rewrites did to the target before picking the calc endpoint, not to apply the remembered fix blindly. A `gcongr` discharged its own nonnegativity side goal, leaving a stray `exact` with no goals. And `rw [Finset.range_eq_Ico]` rewrote both differently-instantiated occurrences at once, so the second copy of the rewrite found nothing; `simp only` expresses the intent (rewrite everywhere) deliberately.

4 Mathlib lookup versus new mathematics

Lookup: `Real.exp_bound` (the Taylor remainder with explicit constant), `Real.abs_exp_sub_one_le`, `Polynomial.eval_eq_sum_range'`, `Finset.sum_Ico_consecutive`, `Asymptotics.isLittleO_pow_pow`, and `pow_le_pow_of_le_one`.

New: nothing mathematical; the observation that the Taylor-truncation composition eliminates the coefficient-identity work is proof engineering, not mathematics — the numbers produced are the same as the consult's recursion.

5 What the next tide inherits

Stage 5 (the numerator expansion): instantiate `exp_graded_expansion` pointwise in z with $a_s = V_s(z)$ and $\rho =$ the scaled remainder of stage 3, integrate over the mesoscopic window against the Gaussian, remove the outer tail with stage 2's calculus, and dominate the coefficient functions $z \mapsto c_j(z)$ (continuity and polynomial growth by induction on the power through `Polynomial.coeff_mul`). If the dominated-convergence step needs the remainder bound in explicit window-uniform form, the quantitative companion of this tide's theorem is the first thing to add — its three pieces are already explicit in the proof.